

Department of Computer Science**Assignment 1****Course Code: CS-151****Name:** _____**Course Title: Database Systems****Registration No.** _____**Total Marks: 20****Discipline/Semester: BSCS 2nd****Date: 24-03-25****Submission Date: 28-3-25**

Instructions:

- **Late submissions will not be accepted.**
- **Copied assignments from any source (including AI) will be marked zero.**
- **Submit assignment in both hard and soft form (i.e. Google classroom).**

**Q-1.
C2]****[GA1, CLO1,*****Explain the following using database concepts:***

- A university is planning to store and manage student records efficiently. How does a data dictionary help in maintaining the structure of the database, and what problems might arise if it is not properly maintained?
- You are developing an e-commerce platform that handles thousands of transactions daily. How would you ensure transaction management follows ACID properties, and why is it important for database reliability?
- A multinational company is expanding its database system to support multiple departments and remote offices. Would you recommend a two-tier or three-tier database architecture? Explain your choice, considering factors like scalability, security, and data independence.